

- 1) Construct three address code for the following:
- if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then $z = x + y * z$
 else $z = z + 1$

soln

Step 1: Condition identification

$(a < b) \text{ and } (c < d) \text{ or } (a > d)$

condition

$a < b$

$c < d$

$a > d$

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STEP 2: Table Creation

L1, L2, L3, L4, L5

STEP 3: TAC for condition

L1:

if $a < b$ goto L2

goto L5

L2:

if $c < d$ goto L4

goto L3

L3:

if $a > d$ goto L4

goto L5

STEP 4: Generate TAC for the "THEN" part

Then addition

L4 → then L5 → else

L4:

$t1 = y * 2$

$t2 = x + t1$

$t2 = z$

goto L6

ELSE:

$z = z + 1$

L5:

$t3 = z + 1$

$z = t3$

L6:

End

FINAL TAC:

L1:

if $a < b$ goto L2

goto L5

L2:

if $c < d$ goto L4

goto L2

L3:

if $a < d$ goto L4

goto L5

L4:

$t1 = y * z$

$t2 = x + t1$

$z = t2$

goto L6:

2. Translat $(a < b)$ and $(c < d)$ or $(e < f)$ into three add Stmt using back patching.

Soln

Step 1: Apply Operator precedence AND has higher precedence then OR, so the expression is interpreted

$[(a < b) \text{ AND } (c < d) \text{ OR } (e < f)]$

Step 2: Generate incomplete jump:

if $a < b$ goto -

goto -

if $c < d$ goto -

goto -

if $e < f$ goto -

goto -

Step 3: HANDLE AND

$(a < b)$ AND $(c < d)$ are true if $a, a < b$ is true
Control must continue to the second condition

Step 4: HANDLE OR

If first AND exp is false the second OR operand $e < f$ must be evaluated

Step 5: Final Backpatch code

1. if $a < b$ goto 3

2. goto 5

3. if $c < d$ goto 7

4. goto 5

5. if $e < f$ goto 7

6. goto 8

7. TRUE

8. FALSE

3. Construct TAC using the foll production for Boolean Expression $(a < b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 || B_2$, $B \rightarrow B_1$, $B \rightarrow |B|$, $B \rightarrow E_1 \text{ relop } t_2$,

$B \rightarrow \text{True}$, $B \rightarrow \text{False}$

Soln

Because this is an OR expression. if any one of the three relation condition is true. the compare expression is TRUE.

check first condition:

L1:

if $a > b$ goto TRUE

goto L2

check the second condition:

L2:

if $c < d$ goto LTRUE

goto L3

check the third condition

L3:

if $e < f$ goto Ltrue

goto LFalse

Final TAC for Q3.

L1:

if $a > b$ goto Ltrue

goto L2

goto L3

L3:

if $e < f$ goto Ltrue

goto Lfalse

Ltrue:

goto LEND

LEND:

4. Generate the three address code for the following program fragment:

while ($a < c$ and $B > d$) do

if $A = 1$ then $C = C + 1$

else

while ($a <= D$) do

$A = A + B$

Soln

L1:

$t1 = C + 1$

$C = t1$

goto L1

L2:

$t2 = A + B$

$A = t2$

L2:

$t3 = A - B$

if $A = 1$ goto L4

goto L5

L4:

$t1 = C + 1$

$C = t1$

goto L1

LB:

if $A <= D$ goto Lb

goto 11

Lb:

$t2 = A + B$

$A = t2$

goto L5

LEND: